



Mohamed Hussain S H

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Experience

Having Sixteen (16+) years of experience in front-end development in the reputed companies like

Presight AI, a G42 Company

Abu Dhabi, as Senior/Lead FrontEnd Engineer (2023 - present)



works as Lead frontend engineer for small microfrontend team(now leading / developing multiple projects with help of AI agents) in Presight AI (a G42 company), working on developing the AI integration projects, data intensive dashboard and data visualization platform for the big data analytics(similar to palantir)

SKY NEWS ARABIA

Abu Dhabi, as Senior FrontEnd Developer (2015 - 2022)



Taking care of complete Frontend Development and performance , SEO , different social media and Ad Integration , Analytics of the site. Led the Frontend Team in Sky News Arabia, architected and developed four greenfield projects in span of two years.

INFOSYS (Client: AT&T)

Chennai, as Frontend Engineer (2010 - 2015)



Developing and Maintaining the web pages for the AT&T , leading telecom operator in USA.

Key Areas and Tech Worked

 HTML, CSS, Javascript, JSON, Canvas, AJAX, SVG, Media and all web APIs.

 Typescript, ReactJS, NEXTJS, React Router, AngularJS(1.x), JQuery

 AI agents SDK, coding agents Claude code, copilot, LLMs and prompt engineering.

 State Management using Redux, Xstate, Mobx, React Query(TanStack Query), SWR.

 Build Tools - Vite, Rspack, esbuild, Webpack, Nx, Monorepo, Turborepo, Gulp, NPM, Yarn(workspaces), PNPM.

 UI Libraries - Mantine, Antd, Storybook, reach UI(headless UI), jQuery UI, Foundation, Bootstrap, Many More

 Kubernetes, Helm, Docker, Cloud Deployment, Git, Gitlab CI/CD, Github Actions, AWS Lambda and S3

 NodeJs Backend, SQLite, Elastic Search, REST APIs, ExpressJS, Linux Shell Scripting, Nginx, PostgreSQL

 JAM stack website with NextJS and Eleventy, also MicroFrontends React Web App(Single-Spa)

🔗 CMS : Strapi NodeJS CMS and custom CMS for Sky News Arabia	🔗 Website Performance Optimization, Browser Internals, Caching, Accessibility and Web Security.	🔗 HTML Media i.e video and players like JW Player, hls.js, Video Streaming.
🔗 Responsive Web Design(RWD), Single Page Application (SPA), Progressive Web Application (PWA), Cross Browser Support	🔗 CSS/Web Animations, MotionUI, HTML-In-Canvas, Greensock, Infinite Scroller(I made the lib with virtualization, dynamic height rows)	🔗 Data Visualizations using D3.js, Echarts, Rechart(react based).
🔗 Search Engine Optimization (SEO), DFP Ads Integration, Google analytics, HTML Email	🔗 TDD using Karma, jest, mocha, Jasmine and Cypress for end user Testing	🔗 Smart TV Application Development for LG(webOS) and Samsung (TizenOS).
🔗 Hybrid App Development Phonegap Cordova	🔗 Browser Automation Using Puppeteer, Web,Chrome extensions	🔗 Figma and Design Basics

Project Experience

- working as Lead frontend Engineer in Developing the data pipeline platform , making the design system and coding agents enablement in the team so that entire team of the BE, Data Engineers can code Frontend using the coding agents like claude code, codex (with some guardrails and skills). Also working on the nodejs backend for the BFF and gitlab CI/CD deployments through helm
- working as Lead Design Engineer developing the design system from Figma on the atomic design system principle and powering the multiple projects across the company. Made the design system AI friendly with storybook MCP Integration along with the coding agent skills to make the adoption easy by the developers and designers.
- Working on multiple analytics dashboards in react for Presight, which is a data intensive application which is used to visualize the data with lot of inner moving states which user can interact and explore the data.
- Very good experience in writing readable/clean/maintainable code (now with coding AI agents/LLMs , have experience in guardrail and reviewing critical parts, putting in place solid foundation, best practices so that coding agents can multiply in the codebase) maintainable semantic HTML, CSS (really good understanding), JavaScript, Typescript and good experience in analyzing the performance issues through dev tools performance panel and memory profiling of JS engine Heap.
- Recently worked on improving the performance improvement of one microapp(issue is like even typing a letter in input box takes some time to appear in UI), there I implemented virtual scrolling (created own library as some features prevent me from using npm libs) and removed the unnecessary re-renders using react-scan (to visualize render) and removed all the deep cloning and made the app to use single reference for data used Ids to access the data and removed unnecessary useEffects and memoizations which improved the overall performance and loading time of the page improved by 50% and it got huge appreciations and clients asked me to share the entire learning to the team and I wrote a detailed blog with what I did and shared.
- In Presight I architected and developed multiple modules, developed multiple share libs with customized requirements(like datepicker) which integrates with elastic search backend , AI services (like streaming data) and microfrontend architecture for the data visualization platform using ReactJS. Recently for the legacy codebase with lot of ts issues, added CI/CD build step(tried with ts-go native port) to prevent introduce new typescript errors into the legacy codebase.
- Architected and developed Sky News Arabia Business website <https://www.snabusiness.com> with ReactJS, NextJS using the APIs. Involved mainly the data visualization using Rechart(D3 variant)
- Client side frameworks: ReactJS (Rehydration with NextJS), AngularJs (with rehydration, I implemented rehydration concept in Angular 1.x) and using jQuery since started and for past five years turning back to plain vanilla js.
- Architected and developed JAM stack bilingual course platform with ReactJS, NextJs, Strapi CMS with Node Backend for Sky news Arabia Academy <https://academy.skynewsarabia.com/en> Multi Language Support(Arabic and English)
- CSS Modules, Styled components (now no css-in-js solutions) antd, mantine, React-Aria (headless ui), reachui (headless ui) and other styling solutions SCSS, LESS and CSS frameworks customized bootstrap and foundation (SCSS), and developed inhouse CSS framework/UI-Lib for Sky News Arabia.

- Experience in working and developing for Strapi CMS, Node JS backend and custom inhouse CMS for Sky News Arabia.
- Editor, now mainly 80% in coding agents CLIs, GUIs and 20% in reviewing critical parts, was primarily Working in VS Code(sublime fan). using coding agents for more than two years now and I am a big fan of coding agents (* with guardrails and skills)
- Experience in working in AWS lambda function and S3 i.e for serving the image using sharp nodejs module and experience in CDN and its workings, cloudflare, cloudfront fastly etc Developed solution to use optimized image using image optization libraries and solutions like sharp, mozjpeg etc.
- Good in understanding the inner working of the render pipeline of the browser and code compilation of the javascript and how the event loop works. I used this in many places to optimize the performance of the web pages.
- Collaborating with designers and business teams in design discussions and delivering solutions on time by having a trade-off in requirements so that feature can reach the customers in time.
- Automating the repeated browser tasks and testing using chrome puppeteer and used cypress for end user testing
- Web performance Optimization using proper caching techniques and browser hints prefetch, preload, service worker and measuring it using lighthouse and pagespeed and webpage test
- Experience in working in RTL(Arabic) Sites, worked in bilingual site (arabic/english)
- Worked in Hybrid phonegap application(sky news football app) and smart TV application for LG and Samssung (target Env is Browser Engine), For TV application I developed a small UI framework with less memory footprint and state management using Xstate.

Education

-  Bachelor of Engineering (DISTINCTION) in Electronics and Communication from Crescent Engineering College, Chennai , India - 2010.

Interests

- Tinkering with latest AI models through the agnostic vercel AI SDK, trying to build multiple AI powered tools and applications and learning about the AI model development and how to leverage the AI models in our applications.
- Learning the internal workings of AI models, GPUs and setup the local AI model inference in my PC(Kimi) and all the AI engineering stuffs, recently impressed with the article on [prompt caching](#).
- Latest web technologies, (now playing with HTML-In-Canvas Apis) and trying to match the web experience with native app like experience with PWA (SW, background sync) and performance optimization for web.
- Learning more about the Server environment especially Linux environment and How the internet works(TCP, DNS, HTTP, UDP Protocols and networking) and more close to the bare metal OS concepts,
- Learning More About distributed systems and system architecture. Learned lot from the book "[Understanding Distributed Systems](#)" by Robert Vitillo and continously following popular ByteByteGo newsletters and articles.
- Following latest trends and techniques of front end in Twitter [@shmdhussain12](#) (I prefer twitter over X), BSky my profile is [@shmdhussain](#) often sharing what I learned/admired/inspired useful things in X
- Doing small cool stuffs, recently setup the linux vm for my personal site and hosted this site, you are looking into this and automating tasks in terminal and learning/deep-diving about inner workings of the tech which I am using.
- Reading books: completed reading of "[AI Engineering](#)" by [Chip Huyen](#) and completed "Fluent React" from Tejas and learned javascript(about 60%) from "professional JavaScript for web developers" by Nicholas Zakas and Dr Axel's "JavaScript For Impatient Programmers"
- Tinkering with browser devtools and latest cool things found is puppeteer scripts through recording user actions in chrome. must follow the [Patrick Brosset's Dev Tool Tips](#) and [Umer Hansa's Dev Tool Tips](#) for cool dev tools stuffs.
- Watching conference talks, watching/reading latest trends in tech space to get to know about trends and latest things in our Ecosystem.

- Learning new tech and trying to dig deep in tech and teaching and sharing with others what I know. Often sharing what I learned/admired/inspired useful things in [@shmdhussain](#)
- Swimming, Playing and watching football and rest of time be with friends and family.

About Me

-  Friendly, kind, smiling, curious, hardworking, lifetime learner.
-  Husband. Father. From Chennai India.

In progress

-  Personal blog with Astro.
-  Making the hands dirty with AI development (no model development, other ways to leverage AI like MCP, RAG and others)